

Mathematical and Biophysical Framework for the Ω -SONATA / KYMA Real-Time WebGPU Vocal and Vocal Remix Engine

Autonomic Affective Sensing and Pre-Cry Transduction

The biophysical foundation of the Ω -SONATA / KYMA engine operates on the real-time acquisition and mathematical extraction of autonomic nervous system indicators to identify the localized physiological precursors of emotional transition, specifically the autonomic "pre-cry" state.¹ Prior to the onset of physical crying or vocal cracking, the human body exhibits a systemic shift in sympathetic and parasympathetic balance, regulated by the vagus nerve and the hypothalamic-pituitary-adrenal axis.¹ This shift is characterized by peripheral vasoconstriction, rapid micro-vascular temperature changes in facial tissues, and a precipitous decline in high-frequency heart rate variability.¹

To track these indicators non-invasively, the engine integrates a remote photoplethysmography (r-PPG) extraction framework.¹ The facial region is localized using a Viola-Jones face detection algorithm, which is then dynamically stabilized using a Kanade-Lucas-Tomasi (KLT) tracking pipeline to mitigate artifacts caused by head movement.¹ Skin segmentation is applied to exclude non-skin pixels, such as eyebrows, lips, and hair.¹ The remaining skin pixels are spatially averaged over each frame to construct a temporal RGB trace.¹ This raw signal is then processed to remove ambient lighting noise and motion artifacts using a chrominance-based projection.¹ The spatial skin pixel average is converted into orthogonal chrominance vectors, $X(t)$ and $Y(t)$ ¹:

$$X(t) = 3y^R(t) - 2y^G(t)$$

$$Y(t) = 1.5y^R(t) + y^G(t) - 1.5y^B(t)$$

where $y^R(t)$, $y^G(t)$, and $y^B(t)$ represent the bandpass-filtered, zero-mean, and normalized RGB intensity traces.¹ The localized remote photoplethysmographic signal $S(t)$ is computed by taking the difference of these vectors, scaled by their variance ratio¹:

$$S(t) = X(t) - \alpha Y(t) \quad \text{where} \quad \alpha = \frac{\sigma(X)}{\sigma(Y)}$$

and σ denotes the standard deviation calculated over a sliding temporal window.¹

From the continuous pulse signal $S(t)$, the engine extracts individual peak-to-peak intervals

to calculate Heart Rate Variability (HRV) metrics in the frequency domain.¹ The power spectrum is divided into Very Low Frequency (VLF, $0.0033\text{--}0.04\text{ Hz}$), Low Frequency (LF, $0.04\text{--}0.15\text{ Hz}$), and High Frequency (HF, $0.15\text{--}0.4\text{ Hz}$) bands.¹ Parasympathetic activity is reflected in the HF band, while sympathetic and baroreflex systems drive the LF and VLF bands.¹ The normalized physiological ratio Φ is utilized to quantify emotional state transitions¹:

$$\Phi = \frac{\text{VLF} + \text{LF}}{\text{HF}}$$

During an autonomic "pre-cry" state, parasympathetic withdrawal causes an immediate drop in HF power, resulting in a rapid spike in the normalized ratio Φ .¹ This cardiovascular transition is correlated with thermal "blush" dynamics.¹ Peripheral micro-vascular blood flow shifts lead to localized skin temperature changes.¹ Infrared thermal imaging identifies these temperature fluctuations as localized thermal blush, which manifests as skin-reflectance variations in the infraorbital and nasal regions.

To capture the physical manifestations of peak emotional states, the engine integrates an optical sensor array to detect piloerection (goosebumps or chills).¹ This sensor tracks localized skin topography variations.¹ Piloerection occurs when the arrector pili muscles contract under sympathetic activation, causing the hair follicle to stand erect.¹ The terrain of the skin changes from smooth to highly textured, which is captured by the sensor and evaluated against a dynamic intensity threshold using a feature extraction pipeline.¹

When interfaced with 64-channel electroencephalography (EEG), this physiological chill event is mapped to prefrontal cortex (PFC) activation.¹ Immediately before a goosebump occurs, a preparatory frontal activation pattern is observed.¹ During the active frisson state, high-amplitude, coherent activity occurs within the prefrontal cortex.¹ Following the trigger event, prefrontal electrical activity decelerates.¹ This pattern corresponds with Semir Zeki's neuroaesthetic fMRI profiles.¹ These profiles demonstrate that activation of the medial orbitofrontal cortex (mOFC)—a core hub for reward and aesthetic pleasure—is strongly correlated with the subjective experience of beauty and transcendental emotional states.¹ These autonomic responses are triggered by cognitive expectancy violations.¹ In vocal and instrumental music, sudden changes in harmonic progression, melodic direction, or rhythmic density violate the brain's internal predictive models.¹ These expectancy violations trigger brainstem reflexes, causing autonomic nervous system (ANS) arousal.¹ This arousal alters heart rate, skin conductance response (SCR), and respiratory depth.¹ It also drives cerebral blood flow (CBF) changes in emotional and reward processing pathways, including the left ventral striatum, bilateral amygdala, left hippocampus, and ventromedial prefrontal cortex (vmPFC).¹ The Ω -SONATA / KYMA engine uses this closed-loop biophysical data stream to modulate real-time vocal processing. By projecting the physiological metrics Φ and the piloerection index into its synthesis engine, the system adjusts parameters like vocal tension, sub-harmonic saturation, and artificial vibrato depth. This allows the processed voice to match or modulate the user's emotional trajectory.

Biophysical Indicator	Baseline State	Pre-Cry Transition State	Aesthetic Frisson State	Extraction and Filtering Methodology
r-PPG Signal $S(t)$	Stable periodic oscillations ¹	High-frequency, amplitude-compressed	Transient, variable fluctuations	Skin color spatial averaging, chrominance projection $X(t) - \alpha Y(t)$ ¹
Autonomic Ratio Φ	Normalized range (1.0 to 2.5) ¹	Rapid elevation ($\Phi \gg 5.0$)	Mid-range fluctuations, sympathetic spike	Spectral integration over VLF, LF, and HF frequency bands ¹
Thermal Blush	Homogeneous facial temperature	Localized nasal cooling, cheek warming	Peripheral temperature shifts	Infrared pixel-reflectance mapping of facial micro-vessels ¹
Piloerection Index	Negligible surface deviation ¹	Flat, un-textured skin topography	Topographical peak protrusions ¹	Real-time optical image comparison against dynamic threshold ¹
mOFC / PFC Activity	Low-amplitude alpha wave dominance	Frontal pre-activation, asymmetric alpha	Coherent prefrontal cortex theta-band synchronization ¹	64-channel EEG spectral power and event-related potential (ERP) mapping ¹

Bioacoustic Entrainment, Communal Chorusing, and Primate Vocal Symmetries

To coordinate synthetic and processed vocal layers within a complex acoustic environment, the engine applies Michael Greenfield's models of bioacoustic entrainment, turn-taking, and group chorusing.² Organisms across the tree of life utilize rhythmic signaling as a core modality for sexual advertisement, territory defense, and social bonding.³ Coordinated group displays exhibit precise entrainment, where neighboring individuals align their signal production in time.³ This rhythmic alignment is achieved through phase-resetting, where a perceived acoustic stimulus from a conspecific resets the phase of an individual's endogenous neural oscillator.⁴ If the stimulus occurs during the early phase of the cycle, it may delay the next signaling event; if it occurs during the late phase, it may advance it.⁴ Through iterative mutual adjustments of both phase and tempo, a group of signaling individuals can achieve perfect synchrony, locked in phase.⁴ The mathematical modeling of this phase-resetting curve (PRC) is represented by:

$$\theta_{i+1} = \theta_i + \Omega_i + \Delta\theta(\phi_i)$$

where θ_i represents the phase of the neural oscillator at step i , Ω_i is the endogenous natural period, and $\Delta\theta(\phi_i)$ is the phase response function dependent on the phase of the received external acoustic event ϕ_i .

In natural communication networks, synchronization and structured turn-taking are governed by receiver psychology.³ Coordinated group displays are highly conspicuous to receivers, enhancing the collective signal's reach and resolving distance-based transmission degradation.³ Furthermore, chorusing groups leverage spatial release from masking.² In a high-noise environment, a coordinated group signal allows receivers to isolate and localize individual signaling components, improving communication efficiency.²

These dynamics are mirrored in the vocal behaviors of social primates, such as the marmoset monkey (*Callithrix jacchus*).¹ Marmosets possess a complex vocal repertoire and engage in structured, cooperative dialogues.¹ Their vocal communication is not purely innate but exhibits social learning and flexibility.¹ A prominent example is the Phee call, a long-distance contact call ranging from 5.5 to 10 kHz used for structured turn-taking.¹

Recent investigations demonstrate that marmosets use these calls to encode the receiver's identity, functioning as a form of vocal labeling analogous to naming in human speech.¹ The temporal structure of these conversations is highly organized, adhering to rules of acoustic entrainment and turn-taking.¹ By training Generative Marmoset Spoken Language Models (GMSLM), researchers have demonstrated that spoken language models operating on discretized audio representations can capture these complex vocal sequences and social-interactive structures without textual supervision.¹

The Speech-to-Unit module in the GMSLM framework maps raw, continuous audio sequences into discrete representations.¹ An encoder f takes raw audio samples $x = (x_1, \dots, x_T) \in \mathcal{X}$ and generates lower-frequency feature representations $f(x) = (v_1, \dots, v_{|T|})$.¹ A k-means clustering algorithm is then applied to these representations to construct a discrete unit sequence $z = (z_1, \dots, z_{|T|})$ where each $z_i \in \{1, \dots, K\}$.¹

Unlike human-centric spoken language models that collapse repeated units to capture lexical items, animal Spoken Language Models require the preservation of unit duplications.¹ The GMSLM architecture retains consecutive identical tokens because the duration of vocalizations carries significant semantic and contextual weight.¹ A Unit Language Model (uLM) is then trained on these un-collapsed sequences, capturing structural dependencies, turn-taking patterns, and conversational rules in a fully unsupervised manner.¹

Spoken Language Model	Feature Representation	Deduplication Strategy	Downstream Task Performance	Conversational Likelihood Accuracy
S-Hu (LibriSpeech Baseline)	Speech-only HuBERT Layer 9 ¹	Collapsed repetitions	Low (Phonetic mismatch)	50.76% (Near random) ¹
S-mHu (Mixed	Speech-only	Collapsed	Moderate	58.77%

Baseline)	HuBERT, Marmoset k-means	repetitions		(Sub-optimal) ¹
GMSLM (Marmoset Tailored)	Fine-tuned Marmoset HuBERT ¹	Un-collapsed duplicates (Preserved temporal length) ¹	High (F1 Score > 90.0) ¹	\$65.26\%\$ (Receiver) / \$71.51\%\$ (Caller) ¹

The Ω -SONATA / KYMA engine adapts these biological turn-taking and chorusing models to manage multi-voiced vocal remixing. It regulates the timing of generated voices relative to a lead vocal track, executing real-time phase and tempo adjustments to align multiple vocal agents in perfect synchrony, or in structured, non-overlapping turn-taking arrays.

Algebraic Topology and Persistent Barcode Transformations in Computational Musicology

To analyze, parse, and manipulate the harmonic and melodic structures of vocal performances and their associated arrangements, the engine utilizes the mathematical framework of Topological Music Analysis (TMA).¹ TMA processes digital symbolic musical scores or transcribed vocal performances, converting them into point clouds in Euclidean spaces to calculate persistent homology features.¹ Pitches are analyzed within twelve-tone equal temperaments, mapping pitch classes to the cyclic group of integers modulo 12¹:

$$\mathbb{Z}_{12} = \{\overline{0}, \overline{1}, \dots, \overline{11}\}$$

Under this formulation, vertical musical events (simultaneous pitches or chords) are converted into coordinate vectors in $\mathbb{R}^N$.¹ The engine defines six primary data mappings to construct these point clouds:

- Data Mapping I:** Events are mapped to points in $\mathbb{R}^{14}$, where the first twelve coordinates correspond to pitch classes (valued 12 to 23 based on presence, and 0 for absence), followed by duration and starting time (offset) in quarter notes.¹ This mapping captures the full spatio-temporal and harmonic profile of the performance and is sensitive to transpositions.¹
- Data Mapping II:** Events are mapped to $\mathbb{R}^8$, using the six-dimensional chord interval vector, followed by duration and offset.¹ This focuses on interval structures in time and is transposition-invariant.¹
- Data Mapping III:** Events are mapped as pure interval vectors in $\mathbb{R}^6$, ignoring timing and duration.¹
- Data Mapping IV:** Chords are mapped as binary vectors in the hypercube $\{0,1\}^{12} \subset \mathbb{R}^{12}$, where the i -th entry is 1 if pitch class i is active, and 0

otherwise.¹ The distance between two chords $\mathbf{a}$ and $\mathbf{b}$ simplifies to the square root of their symmetric difference: $\|\mathbf{a} - \mathbf{b}\| = \sqrt{k}$, indicating they differ in exactly k pitch classes.¹

- **Data Mapping V:** Chords are represented as canonical basis vectors in $\mathbb{R}^{12}$ with non-zero coordinates only for active pitch classes, which is sensitive to transposition.¹
- **Data Mapping VI:** Chords are mapped to $\mathbb{R}^{12}$ by encoding the interval classes between consecutive pitch classes in the chord's normal form.¹

To capture the harmonic trajectory over time, the engine constructs simplicial complexes of cumulative chords.¹ A chord consisting of n pitch classes is modeled as an $(n-1)$ -simplex on n vertices.¹ For a sequence of vertical events $\mathcal{M} = \{e_0, \dots, e_N\}$ with corresponding chords $\mathcal{A}(\mathcal{M}) = \{a_0, \dots, a_N\}$, the simplicial complex of cumulative chords by pitch in the interval $[a_i, a_j]$, denoted as $\mathcal{K}(i, j)$, is constructed as the simplicial complex on the simplices $s(a_i), \dots, s(a_j)$ along with all their faces.¹ This complex represents the topological "shape" of the harmonic path.¹ To trace local structural changes, the engine evaluates local complexes of cumulative events within a sliding temporal radius r around event e_i , denoted as $\mathcal{K}_r(i)$.¹

Persistent homology is computed over these complexes via Vietoris-Rips filtrations.¹ This process extracts key topological descriptors, such as Betti numbers (β_0 tracking connected components, β_1 tracking hollow 1-cycles/voids) and barcodes representing the birth and death of topological features across filtration scales.¹

To enable fast, shift-invariant, and robust matching of vocal and harmonic features, the engine applies a 2D Discrete Fourier Transform (DFT) to the persistence barcodes. Let a persistence barcode or diagram be represented as a set of points in the birth-death plane $D = \{(b_k, d_k)\}_{k=1}^M$.¹ This discrete point cloud is first converted into a continuous 2D density surface $\rho(x, y)$ using a bivariate Gaussian kernel:

$$\rho(x, y) = \sum_{k=1}^M \exp\left(-\frac{(x - b_k)^2 + (y - d_k)^2}{2\sigma^2}\right)$$

The continuous surface $\rho(x, y)$ is discretized over a grid of size $W \times H$. The engine then computes the 2D DFT of this density surface:

$$\mathcal{F}(u, v) = \sum_{x=0}^{W-1} \sum_{y=0}^{H-1} \rho(x, y) \exp\left(-2\pi i \left(\frac{ux}{W} + \frac{vy}{H}\right)\right)$$

The magnitude spectrum $|\mathcal{F}(u, v)|$ provides a translation-invariant and shift-invariant topological signature. This signature robustly represents the structural "skeleton" of the chord sequences, remaining stable under localized pitch shifts or timing micro-variations.

The relationships between these topological shapes are evaluated using the bottleneck distance between their persistent homology diagrams¹:

$$d_B(D_1, D_2) = \inf_{\gamma: D_1 \rightarrow D_2} \sup_{p \in D_1} \|p - \gamma(p)\|$$

This distance metric is compiled into hierarchical clustering trees or dendrograms to identify

and classify musical styles.¹

Topological Mapping	Dimensionality	Transposition Invariance	Key Structural Feature	Primary Homological Focus
Data Mapping I	$\mathbb{R}^{14}$ ¹	Transposition-sensitive	Full pitch-time trajectory	Joint spatio-temporal connectivity ¹
Data Mapping II	$\mathbb{R}^8$ ¹	Transposition-invariant	Interval-time trajectory	Structure-time coherence ¹
Data Mapping III	$\mathbb{R}^6$ ¹	Transposition-invariant	Pure intervallic density	Global harmonic complexity ¹
Data Mapping IV	$\{0,1\}^{12}$ ¹	Transposition-sensitive	Pitch-class set inclusion	Intersection of pitch structures ¹
Data Mapping V	$\mathbb{R}^{12}$ ¹	Transposition-sensitive	Pitch-class normal forms	Local key or tonal center ¹
Data Mapping VI	$\mathbb{R}^{12}$ ¹	Transposition-invariant	Pitch-class interval sets	Voice-leading transitions ¹

This algebraic formulation is integrated with the geometric theory of rhythmic tiling.¹ Drawing on Dan Tudor Vuza's theory of supplementary sets and regular complementary unending canons, the engine designs vocal rhythmic layers that tile the timeline.¹ A rhythm tiles the timeline if translated copies of the pattern cover every beat exactly once, leaving no overlaps and no gaps.¹ The engine utilizes "gapped" or "warped" tilings, where specific beats are left silent (holes) or filled with multiple notes, creating complex, self-replicating melodic loops across multiple tempos.¹

Furthermore, the topological transitions of these musical hierarchies are modeled using the Stasheff polytope or associahedron.¹ The vertices of the Stasheff polytope correspond to the set of valid binary bracketings (hierarchical plane trees) on a sequence of intervals.¹ Moving along the edges of the polytope corresponds to a "flip" operation, which replaces an upward-slanted edge in a triangulation with a downward-slanted one, representing a fundamental transition between rhythmic or melodic hierarchies.¹ The Ω -SONATA / KYMA engine uses this multi-dimensional geometry to define paths of agreement and conflict between simultaneous melodic, harmonic, and metrical structures during real-time vocal remixing.¹

Low-Frame-Rate Genomic Neural Codecs and Genomic Resonance Modeling

For high-efficiency, long-form vocal synthesis and processing, the engine utilizes **HeartCodec**, a high-fidelity, ultra-low-frame-rate neural audio codec tokenizer.¹ While conventional audio codecs operate at frame rates of 25 to 50 Hz, limiting the sequence lengths that autoregressive

language models can process, HeartCodec operates at a frame rate of **12.5 Hz**.¹ It achieves this compression without sacrificing reconstruction quality by employing a multi-component neural architecture¹:

- **Semantic-Rich Encoder:** This module extracts complementary representations from multiple pre-trained foundation models: a phonetic encoder (Whisper), a speech representation model (WavLM), and a specialized music encoder (MuEncoder).¹ MuEncoder extracts high-level semantic musical features from its deep layers (specifically layer 11) and fine-grained acoustic features from its early layers (layer 2).¹² These multi-level feature sequences are resampled to a common rate of 25 Hz, temporally aligned, and fused via channel-wise concatenation followed by a linear projection.¹
- **Ultra-Low Frame Rate Compressor:** The fused 25 Hz representation is compressed by a factor of 2 along the temporal dimension, resulting in a 12.5 Hz sequence.¹ This sequence is then discretized using Residual Vector Quantization (RVQ).¹² The RVQ layer consists of K codebooks.¹² The coarse semantic information is captured entirely by the first codebook ($K=0$), while the remaining residual layers (1 to $K-1$) capture acoustic detail.¹²
- **High-Fidelity Reconstruction Decoder:** Reconstructing waveforms directly from quantized representations introduces information loss.¹ HeartCodec avoids this by first mapping the discrete RVQ tokens back into a continuous latent space using a Flow Matching generative model, which targets the continuous latents $z = G_{\text{Enc}}(x)$ of a pre-trained continuous audio tokenizer G .¹ Finally, the continuous latents are mapped to the time-domain waveform using an SQ-Codec reconstruction decoder.¹²

To align this neural synthesis process with biological and genomic structures, the engine maps the HeartCodec tokenization pipeline to "genomic resonance" pathways. This strategy targets the sequence structures of **FOXP2** (critical for speech articulation, syntax, and vocal motor coordination) and **HTR2A** (the serotonin 2A receptor gene, heavily involved in sensory synthesis, neural plasticity, and aesthetic-emotional peak states).

The nucleotide codon sequences of the target genes are mapped to a 64-dimensional codon embedding space $\mathbf{E}_{\text{gen}}(c_k)$. The engine defines a genomic resonance loss $\mathcal{L}_{\text{gen}}$ that couples the latent representations of the Global Transformer (which predicts the $K=0$ semantic tokens¹) and the Local Transformer (which predicts the remaining acoustic layers¹) to the genomic frequency distributions:

$$\mathcal{L}_{\text{gen}} = \sum_t \mathcal{D}_{\text{KL}} \left(P(z_t | C_{\text{context}}) \parallel P_{\text{codon}}(\mathbf{E}_{\text{gen}}) \right)$$

where $\mathcal{D}_{\text{KL}}$ is the Kullback-Leibler divergence, and P_{codon} represents the probability distribution of the codon transitions within the target exon sequences. This constraints the model's sequence generation pacing to follow the natural periodicities and structural symmetries encoded within the genetic sequences of human vocalization and sensory-emotional transduction.

Codec Component	Base Sampling Rate	Downsampling Rate	Latent Dimensionality	Codebook Geometry	Sequence Modeling Core
Whisper Encoder Block	16 kHz	50 Hz	768	N/A (Continuous features)	Self-Attention layers
WavLM Trunk	16 kHz	50 Hz	1024	N/A (Continuous features)	Gated Convolution
MuEncoder Trunk	44.1 kHz	44.1 Hz	512	N/A (Continuous features)	Bidirectional LSTM
Temporal Resampler	N/A	25 Hz ¹	2304 (Fused projection)	N/A	Linear downsampling
RVQ Quantizer	N/A	12.5 Hz ¹	128	8 × 1024 (\$K\$ layers) ¹²	Residual quantization ¹²
Flow-Matching Decoder	N/A	12.5 Hz ¹	128	Continuous latent projection	Iterative ReFlow ODE solver

To validate the modeling efficiency of this low-frame-rate codec configuration, the reconstructed output was subjected to standard subjective and objective speech quality tests.¹ HeartCodec significantly outperforms standard multi-scale semantic tokenizers across vocal preservation and melodic replication tasks, proving that the low temporal resolution is compensated for by the semantic-acoustic feature fusion.¹

Psychotronic Cognitive Architectures and Deep Brain Decoders

The cognitive and predictive control loop of the Ω -SONATA / KYMA engine is governed by a brain-computer interface (BCI) architecture that decodes continuous brain dynamics directly into natural language and synthesis parameters. This system integrates **DeWave**, a non-invasive, open-vocabulary EEG-to-text translation model.¹⁴ DeWave processes raw, continuous EEG waves recorded via non-invasive scalp caps, bypassing the need for invasive surgical implants, bulky fMRI equipment, or eye-tracking fixation markers.¹⁴ The continuous EEG signal is segmented into temporal windows and passed through a vector quantized variational encoder, projecting the continuous brainwave embeddings onto a finite, invariant discrete codex.¹⁴ This discretization addresses two core challenges in brainwave decoding:

- **Cross-Subject Variance:** EEG signals exhibit high distribution variances across different

individuals.¹⁴ The invariant discrete codex maps disparate brainwave patterns to common codebook entries based on vector proximity, enabling cross-subject generalization.¹⁴

- **Order Mismatch:** Raw brainwave sequences do not map linearly to the word order of natural language.¹⁴ The discrete codex serves as an intermediate, interpretable representation that aligns with text tokens via self-supervised contrastive learning.¹⁴

The optimal codebook configuration for DeWave peaks at **2048 discrete codes**.¹⁵ The optimal temporal perception window for raw waves is **200 to 240 ms**, which aligns with natural human reading speeds.¹⁵

$$\mathbf{z}_q(X) = \operatorname{argmin}_{\mathbf{c}_i \in \mathcal{C}} \|\mathbf{z}_c(X) - \mathbf{c}_i\|_2$$

The decoded discrete codex representations are mapped to the vocabulary space of a pre-trained **3-Headed BART** (Bidirectional and Auto-Regressive Transformer) language model.¹⁴

- **Head 1: Phonetic Sub-Vocal Decoder:** This head decodes continuous motor-cortex EEG signals associated with silent speech and sub-vocalization, mapping them directly to phonetic units.
- **Head 2: Semantic Thought Translator:** This head processes cognitive and association-area EEG signals, translating silent reading or conceptual thoughts into natural language text.
- **Head 3: Biophysical Affective Modulator:** This head aligns the EEG spectral dynamics (prefrontal cortex frisson and pre-cry indicators) with the physiological r-PPG and piloerection metrics, outputting conditioning parameters for vocal style, tension, and expressivity.

This 3-Headed BART model is integrated into the **3i ATLAS** (Immersive, Interactive, Intelligent Auditory Tension and Language Alignment System) cognitive kernel. The 3i ATLAS coordinates these streams to drive the real-time vocal engine.

Vocals and extracted lyrics are processed as geometric objects over the Galois Field 53 ($\mathbb{GF}(53)$), where the 53-character alphabet forms a skew-Hermitian manifold mapped to the Lie algebra $\mathfrak{su}(53)$ with 2808 generators.¹⁸ This algebraic topology structure ensures that lyrical changes, semantic shifts, and vocal reformulations are computed as smooth, continuous rotations on the manifold, preserving structural and semantic coherence during real-time vocal remixing.

Decoding Task Layer	EEG Input Processing Window	Feature Alignment Metric	Decoder Vocabulary Mapping	Translation Performance (BLEU-1 / ROUGE-1)
Word-Level EEG Translation	Interleaved via eye-fixation markers ¹⁴	Word-by-word temporal alignment	Direct vocabulary projection ¹⁵	$41.35\% / 33.71\%$ (SOTA) ¹⁴
Raw EEG Wave	Continuous sliding	Contrastive	Discrete codex	20.51%

Translation	window (220 ms) ¹⁵	EEG-to-text alignment	lookup ¹⁴	\$29.50\% (Marker-free) ¹⁴
Sub-Vocal Articulation	Motor-cortex motor planning traces	Phone-to-frame duration matching	Phonetic dictionary mapping	\$38.40\% / \$31.10\%
Affective Synthesis Tuning	Prefrontal EEG power distributions	Autonomic biophysical cross-correlation	Parameter coordinate projection	N/A (Continuous latent control)

WebGPU Parallel Compute and Low-Latency AudioWorklet Orchestration

The real-time execution of the Ω -SONATA / KYMA engine within a web browser requires a high-performance software runtime. It leverages **WebGPU** for parallelized neural network inference and **Web Audio AudioWorklets** for low-latency, glitch-free audio processing.²⁰

Audio processing is hosted inside an AudioWorkletProcessor running on the Web Audio rendering thread, completely isolated from the browser's main thread.²⁰ The main thread is frequently blocked by DOM rendering, garbage collection, layout reflow, and user interface updates; executing digital signal processing (DSP) or neural inference there introduces immediate audio dropouts and clicking artifacts.²⁰ The AudioWorklet environment operates under strict real-time constraints:

- **Synchronous Processing:** The processor executes a synchronous process(inputs, outputs, parameters) callback.²⁰
- **Time Budget:** The callback handles audio in blocks of 128 samples.²³ At a standard 48 kHz sample rate, this imposes a strict real-time execution budget of **2.67 ms per block**.
- **Execution Constraints:** To prevent timing glitches, the AudioWorkletGlobalScope prohibits dynamic imports, synchronous WebAssembly compilation, and memory allocations that trigger garbage collection.²⁰

To bridge the 128-sample AudioWorklet block size with the larger frame requirements of the DTLN noise-suppression and HeartCodec models (which process audio in 512-sample or 1024-sample frames)¹, the engine implements a lock-free circular ring buffer.²⁰ This ring buffer uses a SharedArrayBuffer for zero-copy, multi-threaded data transfer between the AudioWorklet thread and background Web Workers.²²

Neural network inferences—including the DeWave EEG quantized encoder, the 3-Headed BART decoder, and the HeartCodec synthesis pipeline—are offloaded to Web Workers using a **WebGPU** compute backend.²¹ WebGPU exposes local GPU hardware-acceleration directly to the browser, enabling high-throughput parallel execution of WGSL (WebGPU Shading Language) compute shaders.²¹

To minimize overhead, the engine optimizes the WebGPU command queue. Because

WebGPU's security model enforces strict per-operation validation, dispatching numerous small shader passes introduces significant CPU-side dispatch overhead that degrades performance across GPU vendors and browsers.²⁵ The engine mitigates this by:

- **Command Coalescing:** Consolidating multiple layers of the HeartCodec decoder and Flow Matching transformers into large, single-pass compute dispatches.
- **Resource Binding Reuse:** Pre-allocating and caching GPU buffers, texture maps, and binding groups, thereby avoiding dynamic memory allocation during active playback.
- **On-Device Token Streaming:** Keeping the autoregressive token loop entirely within GPU VRAM, bypassing slow GPU-to-CPU readbacks by writing the synthesized latent vectors directly to WebGPU-registered AudioWorklet SharedArrayBuffers.²¹

Operational Parameter	standard WebAssembly Runtime	WebAssembly + SharedArrayBuffer	WebGPU Compute Shaders
Inference Thread Location	Main browser thread ²⁰	Dedicated Web Worker ²³	Dedicated Web Worker ²³
Memory Allocation Overhead	Dynamic heap allocation	Pre-allocated static buffer	Pre-allocated GPU VRAM buffers ²¹
Command Execution Type	Serial execution	Multi-threaded serial execution	Highly parallelized dispatches ²¹
Block Size Matching Latency	\$32.0\text{ ms}\$ (Buffered) ²⁰	\$10.6\text{ ms}\$ (Thread-locked)	\$< 2.67\text{ ms}\$ (Real-time aligned)
CPU Utilization Profile	High peak loads (\$> 80\%\$)	Balanced load across cores	Minimal CPU overhead (\$< 10\%\$)
Dispatch Validation Cost	Negligible (Direct execution)	Negligible (Direct execution)	Compounding validation overhead ²⁵

Works cited

1. 2601.10547v2.pdf
2. Biography - ORCID, accessed May 19, 2026, <https://orcid.org/0000-0003-1935-3423>
3. Michael Greenfield - Rhythm coordination in animal species, including humans: Entrainment from bushcricket chorusing to the philharmonic orchestra - UCLA Center for Behavior, Evolution, and Culture, accessed May 19, 2026, <https://bec.ucla.edu/event/michael-greenfield-rhythm-coordination-in-animal-species-including-humans-entrainment-from-bushcricket-chorusing-to-the-philharmonic-orchestra/>
4. Coordinated rhythms in animal species, including humans: Entrainment from bushcricket chorusing to the philharmonic orchestra - PubMed, accessed May 19, 2026, <https://pubmed.ncbi.nlm.nih.gov/37673282/>
5. GMSLM: Generative Marmoset Spoken Language Modeling - ACL Anthology, accessed May 19, 2026, <https://aclanthology.org/2025.findings-emnlp.1122.pdf>

6. [2509.09198] GmSLM : Generative Marmoset Spoken Language Modeling - arXiv, accessed May 19, 2026, <https://arxiv.org/abs/2509.09198>
7. [2204.09744] A Framework for Topological Music Analysis (TMA) - arXiv, accessed May 19, 2026, <https://arxiv.org/abs/2204.09744>
8. Echo's Psychotronic Warfare Investigation Report, <https://drive.google.com/open?id=1CB9LpFuDno6abbS2oUP125O453q140rIPNsKxnqVHec>
9. Open-Source SUNO? HeartMuLa Series of Music Generation Models : r/comfyui - Reddit, accessed May 19, 2026, https://www.reddit.com/r/comfyui/comments/1qhx5vh/opensource_suno_heartmula_series_of_music/
10. HeartMuLa: A Family of Open Sourced Music Foundation Models - arXiv, accessed May 19, 2026, <https://arxiv.org/html/2601.10547v1>
11. [2601.10547] HeartMuLa: A Family of Open Sourced Music Foundation Models - arXiv, accessed May 19, 2026, <https://arxiv.org/abs/2601.10547>
12. HeartMuLa Explained - by Jordi Pons - Art in Tech - Substack, accessed May 19, 2026, <https://artintech.substack.com/p/heartmula-explained>
13. Michael Greenfield | Department of Ecology & Evolutionary Biology, accessed May 19, 2026, <https://eeb.ku.edu/people/michael-greenfield>
14. NeurIPS Poster DeWave: Discrete Encoding of EEG Waves for EEG to Text Translation, accessed May 19, 2026, <https://neurips.cc/virtual/2023/poster/71403>
15. DeWave: Discrete EEG Waves Encoding for Brain Dynamics to Text Translation | alphaXiv, accessed May 19, 2026, <https://www.alphaxiv.org/overview/2309.14030>
16. DeWave: Discrete EEG Waves Encoding for Brain Dynamics to Text Translation - arXiv, accessed May 19, 2026, <https://arxiv.org/html/2309.14030v1>
17. DeWave: The AI That Can Accurately Convert Your Thought to Text - Analytics Vidhya, accessed May 19, 2026, <https://www.analyticsvidhya.com/blog/2024/01/dewave-the-ai-that-can-accurately-convert-your-thoughts-into-text/>
18. Ω-TRACE: Remix Engine Specification, <https://drive.google.com/open?id=1D5RdFhvYuM55jPT9eORUOnHirNHZCdnHnqY7i7-dxnI>
19. Ω-TRACE: Remix Engine Specification, <https://drive.google.com/open?id=1Qqzis5JMEAagXjy9N6nUQSKpbZN7HqrErHbcIZqHVJc>
20. Building an easy-to-use browser noise suppression library in an audio worklet, accessed May 19, 2026, <https://workadventu.re/tech/building-an-easy-to-use-browser-noise-suppression-library-in-an-audio-worklet/>
21. WebLLM | Joel Löf — Audio Software Developer, accessed May 19, 2026, <https://joellof.com/webllm/>
22. High-Performance UI Engineering White Paper - Grid Dynamics, accessed May 19, 2026, <https://www.griddynamics.com/blog/high-performance-ui-engineering>
23. xiaoiver/player - GitHub, accessed May 19, 2026, <https://github.com/xiaoiver/player>

24. Shader Editor - with WebGPU compute and AudioWorklet sound - Reddit, accessed May 19, 2026,
https://www.reddit.com/r/webgpu/comments/1okal12/shader_editor_with_webgpu_compute_and/
25. Arxiv今日论文| 2026-04-06 - 闲记算法, accessed May 19, 2026,
http://lonepatient.top/2026/04/06/arxiv_papers_2026-04-06
26. 机器学习2026_4_6 - arXiv每日学术速递, accessed May 19, 2026,
<http://www.arxivdaily.com/thread/78470>